

Xin Liu

Curriculum Vitae

Department of Electrical and Computer Engineering (ECE)
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Computational Optics; Optical Simulation; Light Shaping; Computational Imaging
— I develop algorithms to simulate and shape light for imaging and sensing.

Education and Experience

- 04/2024 – Present **Postdoctoral Researcher**, Department of Electrical and Computer Engineering, The University of Hong Kong (HKU), Hong Kong SAR, China.
- 01/2024 – 03/2024 **Research Assistant**, State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU), Hangzhou, China.
- 09/2018 – 12/2023 **Ph.D., Optical Engineering**, State Key Laboratory of Extreme Photonics and Instrumentation, Zhejiang University (ZJU), Hangzhou, China.
Thesis: Research on principles and key issues of point spread function engineering
- 09/2014 – 06/2018 **B.E., Optoelectronic Information Science and Engineering**, South China University of Technology (SCUT), Guangzhou, China.

Journal Publications (selected)

(*Equal contributors; ✉Corresponding authors)

- [11] **Retained accuracy with reduced precision in wave propagation modeling.** Xin Liu and Yifan Peng. 2026, *Photonics Research*.
- [10] **Learned off-aperture encoding for wide field-of-view RGBD imaging.** Haoyu Wei, Xin Liu, Yuhui Liu, Qiang Fu, Wolfgang Heidrich, Edmund Y. Lam, and Yifan Peng. 2025, *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*.
- [9] **In situ fully vectorial tomography and pupil function retrieval of tightly focused fields.** Xin Liu, Shijie Tu, Yiwen Hu, Yifan Peng, Yubing Han, Cuifang Kuang, Xu Liu, and Xiang Hao. 2025, *Nature Communications*.
- [8] **Diffraction modeling between arbitrary planes using angular spectrum rearrangement.** Yiwen Hu*, Xin Liu*✉, Xu Liu, and Xiang Hao✉. 2025, *Optica*. [Monthly Top Downloads in January]
- [7] **Modeling off-axis diffraction with the least-sampling angular spectrum method.** Haoyu Wei*, Xin Liu*, Xiang Hao, Edmund Y. Lam, and Yifan Peng. 2023, *Optica*. [Monthly Top Downloads in July & August]
- [6] **Fast generation of arbitrary optical focus array.** Xin Liu, Yiwen Hu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao. 2023, *Optics and Lasers in Engineering*.
- [5] **Investigating deep optics model representation in affecting resolved all-in-focus image quality and depth estimation fidelity.** Xin Liu*, Linpei Li*, Xu Liu, Xiang Hao, and Yifan Peng. 2022, *Optics Express*.
- [4] **Calibration of phase-only liquid-crystal spatial light modulators by diffractogram analysis.** Xin Liu, Shijie Tu, Cuifang Kuang, Xu Liu, and Xiang Hao. 2022, *Optics and Lasers in Engineering*.

- [3] **Accurate background reduction in adaptive optical three-dimensional stimulated emission depletion nanoscopy by dynamic phase switching.** Shijie Tu, [Xin Liu](#), Difu Yuan, Wenli Tao, Yubing Han, Yan Shi, Yanghui Li, Cuifang Kuang, Xu Liu, Yufeng Yao, Yesheng Xu, and Xiang Hao. 2022, *ACS Photonics*. **[On the Cover]**
- [2] **Generation of arbitrary longitudinal polarization vortices by pupil function manipulation.** [Xin Liu](#), Yifan Peng, Shijie Tu, Jun Guan, Cuifang Kuang, Xu Liu, and Xiang Hao. 2021, *Advanced Photonics Research*. **[On the Cover]**
- [1] **Aberrations in structured illumination microscopy: A theoretical analysis.** [Xin Liu](#), Shijie Tu, Yan Xu, Hongya Song, Wenjie Liu, Qiulan Liu, Cuifang Kuang, Xu Liu, and Xiang Hao. 2020, *Frontiers in Physics*.

Awards

- 2025 **Outstanding Young Rising Star Awards (1st/20)**, *The 5th International Conference on Computational Imaging (CITA 2025)*, Chinese Society for Optical Engineering.
- 2024 **Honorable Mention**, *The 4th International Conference on Computational Imaging (CITA 2024)*, Chinese Society for Optical Engineering.
- 2022 **Individual Scholarship for innovation and entrepreneurship**, *Zhejiang University*.
- 2021, 2022 **Merit Graduate Student**, *Zhejiang University*.
- 2020, 2021, 2022 **Junshi Chen Scholarship**, *Zhejiang University*.
- 2020, 2021 **Outstanding Graduate Student Cadres**, *Zhejiang University*.
- 2019, 2021, 2022 **Outstanding Graduate Student**, *Zhejiang University*.

Invited Talks

- 09/2025 **Exploring advanced imaging with computational optics**, *The 5th International Conference on Computational Imaging (CITA 2025)*, Suzhou, China.
- 09/2024 **Free-space wave propagation modeling with least-sampling angular spectrum method**, *The 4th International Conference on Computational Imaging (CITA 2024)*, Xiamen, China.
- 05/2024 **Exploring wave propagation modeling in free space**, *Guest Lecture*, King Abdullah University of Science and Technology, Saudi Arabia.

Student Mentorship

- 2022 – 2025 **PhD Thesis**, “*Learned wide-angle imaging with optical modeling and optimization*”, Haoyu Wei, The University of Hong Kong.
- 2021 – 2025 **PhD Thesis**, “*Characterization of optical field with computational methods*”, Yiwen Hu, Zhejiang University.
- 2023 **Master’s Thesis**, “*Deep stereo optics imaging for high fidelity depth estimation*”, Yuhui Liu, The University of Hong Kong.
- 2022 **Bachelor’s Thesis**, “*Full vectorial manipulation of tightly focused optical field*”, Haiwei Wang, Zhejiang University.
- 2022 **Bachelor’s Thesis**, “*Numerical simulation of vectorial field based on vectorial diffraction theory*”, Yunpeng Wang, Changchun University of Science and Technology.

Service

- Reviewer** Optica; SIGGRAPH (Asia); Advanced Functional Materials; International Conference on Computational Photography (ICCP); Laser & Photonics Reviews; Advanced Science; Optics Letters; Optics Express; JOSA A; JOSA B; Applied Optics; Journal of Microscopy; Optics Communications

Area Chair International Conference on Computational Photography (ICCP), 2026
Committee The 5th International Conference on Computational Imaging (CITA), 2025
Guest Editor Computational Methods for Advanced Optical Imaging, Photonics, MDPI, 2026
Coordinator HKU-KAUST Joint Postgraduate Workshop on Computational Imaging, 2025

Patents

- 2022 **A method and device for calibration of phase-only spatial light modulators**, *Xiang Hao, Xin Liu, Yubing Han, Cuifang Kuang, Xu Liu*, CN112697401A.
- 2020 **A method for point spread function reconstruction**, *Xiang Hao, Xin Liu, Cuifang Kuang, Xu Liu*, CN110133849B.
- 2020 **A two-dimensional rapid beam scanning module for confocal fluorescence microscopy**, *Xiang Hao, Xin Liu, Shijie Tu, Xu Liu*, CN109491065A.
- 2020 **A device for common-path beam modulation in imaging and lithography systems**, *Xu Liu, Xin Liu, Xiang Hao, Cuifang Kuang, Haifeng Li*, CN110568650B.
- 2020 **A device for generating super-resolution optical focus arrays**, *Xiang Hao, Xin Liu, Cuifang Kuang, Xu Liu*, CN110568731B.
- 2020 **A multicolor super-resolution microscope with automatic alignment capability**, *Xiang Hao, Shijie Tu, Xin Liu, Xu Liu*, CN109358030B.